BUSINESS CASE · YTL HACKATHON 2026, TRACK A

Udyam Digital: small businesses go digital in one sitting.

A phone-first, multilingual journey that lets a small business owner find the right bundle, buy it and get it live with minimal human effort. AI handles acquisition; people handle exceptions.

Accel Telecom and Accel Business are the case's fictional operator. All data is synthetic.

01 THE PROBLEM

Buying business connectivity takes a visit, a slow quote, paper KYC and 17.6 days.

7.4%

of leads become customers

368 of 5,000 · Apr–Jun 2026

17.6 d

from lead to live

9.0 of them before a quote is accepted

Rs 15,053

to acquire one customer

2.2× the median monthly bill

3.0

status-chase calls each

6-month churn climbs 6% → 28% with chasing



The quote is the bottleneck. Only 43.6% of assessed leads accept it.

Speed beats price. 41.2% of drops are about reach and speed; 23.9% about price; only 5.9% about paperwork.

Rules are broken. 56 of 368 customers hold bundles the catalogue forbids.

Source: case dataset, smb_leads.csv (5,000 leads)

02 TARGET CUSTOMER

Launch with professional offices; keep micro shops self-serve.

SEGMENT	SHARE OF LEADS	CONVERTS	MEDIAN BUNDLE / MO	6-MO CHURN	CAC	HAS GSTIN	PLAN
Professional offices	**24.9%**	**11.1%**	**Rs 6,575**	**13.8%**	**Rs 10,695**	**86%**	**Launch segment**
Small multi-site businesses	15.1%	11.0%	Rs 28,080	10.8%	Rs 15,614	98%	Assisted mode
Small manufacturers	5.8%	12.8%	Rs 15,910	8.1%	Rs 15,324	99%	Assisted mode
Micro retail & services	54.2%	4.1%	Rs 2,470	22.7%	Rs 20,007	53%	Self-serve wave 2

Who they are

Law offices, coaching centres, travel agencies, diagnostic labs. A median of 14 staff on one site.

Why first

The lowest acquisition cost, strong conversion, and 86% have a GSTIN, so KYC can be pre-filled.

What they want

Faster fibre, endpoint security, a toll-free number, cloud email: bundles worth about Rs 6,600 a month.

Source: case dataset; analysis/05_report_facts.md

03 VALUE FOR THE OWNER

From a visit and 17.6 days to one session on a phone, in their own language.

DISCOVER

Six questions

An advisor in 8 Indian languages, by voice or tap. The GSTIN is checked live at question 3.

BUY

One monthly price

A reason for every item, add-ons that toggle, and a line comparing it with what they pay today.

KYC

Checked on upload

Consent first. Registry lookups pre-fill details; 13 checks catch a stale bill now, not days later.

GO LIVE

Dates, not chasing

Install books itself. A tracker and WhatsApp updates; a named person calls if anything runs late.

8 days

target, lead to live

The owner never waits for a callback or a PDF quote. Eligibility, quantities and prices come from a rules engine, so **no one is sold a bundle they can't use**. The AI only words the explanation.

Shown working in the prototype and the 2:40 demo video

04 VALUE FOR ACCEL BUSINESS

Five pilot metrics, with baselines recomputed from the data.

METRIC	TODAY		PILOT TARGET	WHAT MOVES IT
Lead-to-activation	7.4%	→	**11%**	Instant plan and price; drops at the quote step recovered by self-serve
Cost per activation	Rs 15,053	→	**Rs 8,000**	Discovery and KYC move to the app; assisted deals about 40% cheaper
Days, lead to live	17.6	→	**8**	No visit, no PDF quote; documents verified at upload; install auto-booked
Status-chase calls	3.0	→	**0.5**	Tracker with expected dates; WhatsApp updates; automatic escalation
6-month churn	15.2%	→	**11%**	Less chasing (churn is 6% at 1 call, 28% at 5+) and right-sized bundles
Ineligible bundles sold	**56**	→	**0**	**The rules engine has the final say; tested in 15 judge-style scenarios**

The leads dashboard computes all five live from the database, so the pilot is measured the same way the baseline was.

Baselines: smb_leads.csv. Targets: team estimates to validate in the pilot

05 PRICING

One transparent price, set by rule, inside the catalogue's range. No discounts.

MICRO_MIN_V1

Micro retail & services

Catalogue minimum. The segment is price-sensitive: its median bundle is Rs 2,470 and "price too high" is the third-largest drop reason (414 leads).

MIDPOINT_V1

Offices, multi-site, manufacturers

Rounded midpoint of min and max: a fair middle price that leaves room for the sales team on assisted deals.

EXAMPLE BUNDLE	RULE	MONTHLY	ONE-TIME SETUP	CONTRACT
Kirana, 3 staff, no GSTIN: Biz Fibre 100 + 2 SIMs + Cloud POS + CCTV	micro_min_v1	Rs 2,395	Rs 4,300	12 months
Clinic, 12 staff: Office Pro core	midpoint_v1	Rs 6,055	Rs 0	12 months
Same clinic + Static IP + 1 more email seat	midpoint_v1	Rs 6,443	Rs 0	12 months

Monthly total, setup and contract are shown separately. Bundles cost a median 2.6× the owner's current spend, so the quote says plainly that the plan includes services they don't have today.

Source: docs/PRICING.md; backend/app/pricing/service.py (tested for all 25 products)

06 SAVINGS AND REVENUE

Per 5,000 leads a quarter, at the pilot targets.

+180

activations per quarter

550 vs 370 today

Rs 11.7 L

less acquisition spend / quarter

Rs 44.0 lakh vs Rs 55.7 lakh today

Rs 1.46 cr

added yearly revenue

176 more customers paying at 6 months × Rs 6,900

835

fewer chase calls / quarter

275 vs 1,110 today

How it's calculated

- Activations = leads × conversion (7.4% → 11%)
- Spend = activations × cost per activation (Rs 15,053 → Rs 8,000)
- Revenue = extra customers still paying at 6 months (churn 15.2% → 11%) × Rs 6,900 median bill × 12

Why we believe it

- 32% of drops are directly addressable by self-serve, 69% including partly
- Leads with a GSTIN activate at 9.6% vs 2.0%; the advisor asks for it early
- Run your own numbers: the calculator on the Impact slide of the website

Same formulas as the website's impact calculator; targets to validate in the pilot

07 ROLLOUT

Pilot narrow, prove the five metrics, then scale to the field force and micro shops.

0–3 MONTHS · PILOT

- Professional offices in 2 tier-1 cities
- Real GSTN / Udyam lookups through a GSP or KYC vendor
- DigiLocker pull for documents; Aadhaar e-sign for the order form
- Name a grievance officer; publish the privacy notice

3–6 MONTHS · SCALE

- WhatsApp Business API in production; real SMS OTP
- Postgres with keys in a cloud KMS; key rotation
- Partner and field-force portal: scored leads, assisted mode, shared credit
- Micro-retail wave with a lean bundle

6+ MONTHS · LATER

- Plug into a DPDP consent manager
- Voice-first flow on WhatsApp
- Retrain propensity on live outcomes; monitor bias by segment and language
- Single bill and one support number across products

Go / no-go after the pilot: conversion ≥ 11%, cost per activation ≤ Rs 8,000, KYC completion 68.3% → 85%, first response to escalations within 4 business hours.

The field force is part of the plan: the app hands them scored leads, not fewer deals

08 RISKS

What could go wrong, and what is already built to stop it.

RISK	WHAT COULD HAPPEN	MITIGATION
AI hallucination	The advisor invents a product, price or promise	The AI only explains rules-engine output; product IDs and prices come from code
Wrong recommendation	An ineligible or oversized bundle	Rules engine with 31 automated checks; 15 judge scenarios as the regression test
OCR errors	Raw OCR read only 1 of 11 GSTINs exactly	Checksum and registry repair; the owner types the GSTIN, OCR cross-checks; low confidence goes to review
Privacy and consent	KYC data without valid consent; IDs leak	Consent before upload; minimum data; masked IDs; encryption; auto-delete; audit log
External providers fail	Registry, voice or WhatsApp outage	Provider interfaces with fallbacks: manual entry, text chat, in-app tracker, SMS
Agents overloaded	Escalations pile up	Defined triggers only; least-loaded, language-matched assignment; response time on the dashboard
Synthetic data	Patterns are planted, not observed in the market	Treat findings as directional; the pilot validates every target before scale
Field-force pushback	Sales see the app as a threat	Scored leads, assisted mode and shared credit for app-closed deals

Full register: Phase 1 report, Section 18

09 THE ASK

A plan in minutes. A target of 8 days to live, not 17.6.

Proven in the build

- ✓ One-session journey on a phone, language to live, with real uploads
- ✓ Zero ineligible bundles: the rules engine has the final say
- ✓ Bad documents caught at upload with a specific fix (12/12 KYC verdicts)
- ✓ Late steps escalate on their own to the right agent
- ✓ 192 automated tests and 15 judge scenarios pass

To validate in the pilot

- ? How many drops self-serve recovers (32% direct to 69% including partly)
- ? A 50/50 self-serve / assisted mix, with assisted deals 40% cheaper
- ? KYC completion 68.3% → 85%
- ? Whether regional-language voice closes the quote gap

Decision requested: approve a 3-month pilot with professional offices in two tier-1 cities, measured on the five metrics, with real registry lookups through a KYC vendor.